



Postgraduate Tribology Conference

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**1-2 July 2026**



University of  
**Sheffield**

## Welcome to TriboUK 2026

[Placeholder — the welcome address from the TriboUK 2026 organising committee will be added here.]

## DAY 1 — WEDNESDAY 1 JULY

### MAPPIN HALL

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 09:00–09:30 | Registration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 09:30–09:45 | Welcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 09:45–10:15 | <b>Keynote 1 — Eladio Hurtado Molina</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 10:15–11:15 | <b>Presentation Session 1</b><br><b>Yun Zhao</b> · Imperial College London<br>Lubricating behaviour under electrified conditions<br><b>Zhifeng Hu</b> · University of Huddersfield<br>Condition monitoring of lubricant degradation in journal bearing system based on wireless on-rotor sensing method<br><b>Michael Bartram</b> · University of Sheffield<br>An investigation into the efficacy of relubrication cycles in reducing wear in grease lubricated pitch bearings                                                  |
| 11:15–11:30 | Coffee break<br><i>Sponsored by Phoenix Tribology</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 11:30–12:30 | <b>Presentation Session 2</b><br><b>Sofia Mushtaq</b> · University of Sheffield<br>Measuring forces experienced on human skin during frictional interactions through a combined approach of digital image correlation (DIC) based strain mapping and machine learning<br><b>Zhen Dong</b> · Imperial College London<br>Laser surface texturing of surgical blades<br><b>Osian Thomas</b> · Ingram Tribology<br>Investigation into the tribological differences between plant and animal proteins using a novel test methodology |
| 12:30–13:10 | Travel to AMRC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### AMRC

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|-------------|-------------------------------|
| 13:10–13:50 | Lunch                         |
| 13:50–14:00 | Introduction to AMRC          |
| 14:00–14:30 | <b>Keynote 2 — Leon Proud</b> |
| 14:30–15:30 | Lab tours                     |
| 15:30–15:55 | Networking & closing remarks  |
| 15:55–16:40 | Travel to University          |

### MAPPIN HALL

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| 16:40–17:50 | Drinks reception<br><i>Sponsored by PCS Instruments</i> |
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### GUYSHI

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| 19:00–23:00 | Conference dinner |
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## DAY 2 — THURSDAY 2 JULY

### MAPPIN HALL

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| <b>09:00–09:30</b> | Arrival & refreshments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>09:30–10:00</b> | <b>Keynote 3 — Roger Lewis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>10:00–11:00</b> | <b>Presentation Session 3</b><br><b>Song Yang</b> · Imperial College London<br>Advanced acoustic testing to lubricated contacts<br><b>Oluwatamilore Adenipekun</b> · University of Sheffield<br>An alternative to leaf layer contamination for railhead cleaning trials<br><b>Siyu Wang</b> · Loughborough University<br>Tribological characterization of oil lubricated EV bearing under an electric field                                                                                                                                                                                                                                                                |
| <b>11:00–11:15</b> | Coffee break<br><i>Sponsored by Phoenix Tribology</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>11:15–12:35</b> | <b>Presentation Session 4</b><br><b>Ziyuan Ren</b> · Loughborough University<br>The influence of nonlinear friction on transmission error and torsional stiffness in harmonic reducers<br><b>Seona Mauchline</b> · University of Birmingham<br>Interfacial mesh asymmetry governs lubrication and frictional transition in cartilage biomimetic hydrogels<br><b>Charlotte Currie</b> · University of Sheffield<br>Understanding the wear of ceramic abradable liners in jet engine turbine seals<br><b>Paula Sebastian Asenjo</b> · Brunel University of London<br>Optimising radial air foil bearing performance: balancing clearance and stiffness for enhanced lift-off |
| <b>12:35–13:00</b> | Poster flash talks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>13:00–14:00</b> | Lunch & poster session                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>14:00–14:30</b> | <b>Keynote 4 — Klaus-Dieter Meck</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>14:30–15:30</b> | <b>Presentation Session 5</b><br><b>Musab Rizwan Kazi</b> · University of Lancashire<br>Additive friction stir deposition (AFSD) for light weight alloys<br><b>Uresha Madhuwanthi Herath Mudiyansele</b> · Sheffield Hallam University<br>Slurry erosion and erosion-corrosion response of niobium as an alternative cladding material for neutron-producing targets at ISIS<br><b>Faraz Shaikh</b> · University of Leeds<br>Investigation of friction, wear and particle emissions from coated and uncoated commercial vehicle brake rotors using a small-scale tribological approach                                                                                     |
| <b>15:30–15:35</b> | Voting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>15:35–16:00</b> | Coffee break<br><i>Sponsored by Phoenix Tribology</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>16:00–16:30</b> | Awards & close                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Keynote Speakers

### KEYNOTE 1

#### **Eladio Hurtado Molina**

*Lead Engineer, Blade Bearings — Vestas*

Eladio Hurtado Molina is a leading expert in wind turbine component reliability, currently serving as Lead Engineer for Blade Bearings at Vestas. After earning his PhD from the University of Sheffield in 2022, he successfully transitioned from small-scale wear research to validating components for the world's largest multi-MW wind turbines. Today, he continues to bridge the gap between industry and academia by managing a strategic collaboration between Vestas and the University of Sheffield.

### KEYNOTE 2

#### **Leon Proud**

*Advanced Manufacturing Research Centre (AMRC)*

### KEYNOTE 3

#### **Roger Lewis**

*Professor of Mechanical Engineering, University of Sheffield*

Professor Roger Lewis is a Royal Academy of Engineering Research Chair in the Department of Mechanical Engineering at the University of Sheffield, where he has been an academic since 2002. His research spans industrial wear problems, the development of novel ultrasonic techniques for machine element contact analysis, and the design of engineering components and machines. His work covers wheel/rail contact tribology, friction management, condition monitoring, and the tribology of human interactions including skin friction, hand/object contact, and pedestrian slips. He has been recognised by the Tribology Trust Bronze Medal (2001), a Royal Society Brian Mercer Award for Innovation (2003), the Institute of Physics Innovation in Tribology Prize (2008), and the Donald Julius Groen Prize in Tribology (2020).

### KEYNOTE 4

#### **Klaus-Dieter Meck**

*Innovation Director, Core Technology & Simulation — John Crane*

Klaus-Dieter Meck is the Innovation Director for Core Technology & Simulation at John Crane. Mr Meck has over 32 years of experience in sealing technology and rotating equipment. He is also a named inventor on patents related to mechanical seals, power transmission couplings, and condition monitoring. Mr Meck has recently been appointed as a Visiting Professor to the School of Mechanical, Aerospace & Civil Engineering at the University of Sheffield.

## Poster Presentations

**Alvaro Barrueto** · University of Sheffield

How to replenish lubricant in an oscillating wind turbine blade bearing raceway?

**Chenxi Wang** · University of Leeds

Microstructure-controlled mixed lubrication by WPI-stabilised water-in-water emulsions

**Damien Dooley** · University of Sheffield

The future of dry gas seal materials

**MM Raihan** · University of Leeds

Real-time wear-corrosion measurement using eddy current and electrochemistry sensing

**Mingxin Wang** · University of Leeds

Minimised delubrication of legume proteins by enzymatic hydrolysis

**Mu Qiao** · University of Leeds

The effect of lubricant contamination on fretting wear of engine piston rings and cylinder liners in PHEVs

**Soumya Sikdar** · University of Leeds

Development of PFAS-free ionic liquid greases for tribological applications

**Tianhao Xie** · University of Sheffield

Evaluating the friction recovery and damage behaviour of sand alternatives in the wheel-rail contact

**Yuyang Yuan** · University of Leeds

Fabrication of photoluminescent thin films via tribochemical deposition of upconversion nanoparticles

**Zizhao Wu** · University of Sheffield

Probabilistic distribution of ball-raceway contact force and contact pressure in the blade pitch bearing of the IWT7.5 reference wind turbine

## Our Sponsors & Partners

TriboUK 2026 is made possible through the generous support of our sponsors and partners. We are grateful for their commitment to the tribology community and to early-career researchers.







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